

Composition of linear functions

Input file: **standard input**
Output file: **standard output**
Time limit: 5 seconds
Memory limit: 256 megabytes

An inventor is working to remake the drum machine, an instrument that changed the sound of the music scene around the globe. He's basing his work on a Roland 808 2.0 drum machine.

As he works to create the basic sounds for his new instrument, he has to sort through a lot of different options. At one point, he realizes that he had an ideal version of the basic sound a few iterations ago. Now he needs to get that sound back, only he can't remember exactly what it was or how long ago he heard it.

Help the inventor find the right sound for his drum machine by going through all the options he already found sequentially in reverse order.

Since you're an expert in algorithms, try to imagine the problem like this: elements must be removed one by one from an ordered set of linear functions. For each element, solve the equation $f(g(h(...(x)))) \equiv 0 \pmod{10^9+7}$ (it's this formula that determines the ideal basic sound) in integers, where all other currently remaining functions are in the composition in the order they follow.

Input

The first line contains single positive integer $2 \leq n \leq 10^5$ — the initial number of linear functions in the set.

Next n lines contains the coefficients of linear functions — $i+1$ line contains two integers $0 \leq a_i, b_i < 10^9+7$, where $f_i(x) = a_i x + b_i$.

Next line contains $n - 1$ integers $c_1, \dots, c_{n-1}$, c_i is the number of the function to be removed among remaining functions, and it's guaranteed that $1 \leq c_i \leq n + 1 - i$.

Output

n lines, each of which contains one integer $0 \leq d_i < 10^9 + 7$ — the root of the equation $F_i(x) \equiv 0 \pmod{10^9 + 7}$, where $F_i(x)$ is the composition of all currently remaining functions in the order they follow, starting with $F_1(x) = f_1(f_2(...(f_n(x))))$, the composition of all n functions, and ending with $F_n(x)$, the last remaining function. If the equation has no roots or more than one suitable root, output -1 in the corresponding line.

Examples

standard input	standard output
3 1 1 1 2 1 3 1 2	1000000001 1000000002 1000000005
3 2 1 3 2 4 1 1 1	541666670 583333337 750000005